

Gaspard BEAUDOUIN

Machine Learning research student

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Research interests: World Models, Flow Matching, Representation Learning, VLMs, Masked Diffusion Language Models ...

EXPERIENCE

Incoming Research Scientist Intern — Mistral AI

Paris, France

Multimodal team

2026

- Conduct research on Vision Language Models, representation learning and image encoders.

Research Intern — Harvard University

Boston, MA, USA

Harvard AI and Robotics Lab, MEE

Feb 2025 – Aug 2025

- Conducted research under Dr. M. Wang on generative modelling and flow matching. Proposed DVRF, an inversion-free flow matching framework for text-to-image editing (*CVPR 2026 under review*), and Co-authored *SplitFlow (NeurIPS 2025)* for flow aggregation.
- Co-authored *PAGE-4D (ICLR 2026)* for pose and geometry estimation of dynamic scenes.

Machine Learning Engineer Intern — Sinequa

Paris, France

Research Team

Aug 2024 – Jan 2025

- Studied LLMs and tool-use agents (SFT and RL); designed evaluation for function-calling models (BFCL), data augmentation and templating.
- Distributed instruction fine-tuning (LoRA and full) and DPO on H100 clusters for tool-use; Docker and Slurm.

PUBLICATIONS

Delta Velocity Rectified Flow for Text-to-Image Editing *Accepted at CVPR 2026, Preprint: 2509.05342.*

G. Beaudouin, M. Li, J. Kim, S.-H. Yoon, M. Wang

PAGE-4D: Disentangled Pose and Geometry for 4D Perception *Accepted at ICLR 2026, Preprint 2510.17568.*

K. Zhou, Y. Wang, G. Chen, G. Beaudouin, F. Zhan, P. Liang, M. Wang

SplitFlow: Flow Decomposition for Inversion-Free Editing *Accepted at NeurIPS 2025.*

S.-H. Yoon, M. Li, G. Beaudouin, C. Wen, M. Azhar, M. Wang

EDUCATION

École Normale Supérieure Paris-Saclay

Paris, France

MVA Master — Mathematics, Vision, Learning

2025 – 2026

- Coursework: Advanced learning for text/graphs, DL, Optimal Transport, Probabilistic Graphical Models, Large-Scale LLM Training, Generative Modelling, Representation Learning, Stochastic Calculus, RL, LLM for code and proof.

École Nationale des Ponts et Chaussées — Institut Polytechnique de Paris

Paris, France

Mathematics and Computer Science Department

2022 – 2026

- Focus: Deep Learning, Machine Learning, Computer Vision, Statistics, Convex Optimization, Stochastic Processes, Advanced Programming, Control Theory and Functional Analysis.

Fénelon Sainte-Marie

Paris, France

Preparatory Classes in Mathematics and Physics

2020 – 2022

- 1st Year: PCSI; 2nd Year: PSI* (Physics, Mathematics, Engineering Sciences).

PROJECTS

LeJEPA Re-implementation and Analysis

Master Research Project

Re-implemented LeJEPA, using a Tiny ViT implemented from scratch in PyTorch (17M); studied training dynamics, online linear probing behavior.

2025

GPT MoE Transformer from Scratch

Personal Project

Implemented and trained a PyTorch MoE Transformer in order to generate French poetry with some GPT-OSS recent techniques. I also generated some synthetic training data, and wrote my notes on LLMs: From Pre-Training to RLHF (Draft)

2025

Mini GRPO

Personal Project

Reimplemented GRPO, from by DeepSeek-R1 and DeepSeekMath, applied to math reasoning tasks.

2025

Drug Design with Diffusion Models

École des Ponts

Implemented DDPM, explored DiffDock for drug binding discovery (with Sanofi). Poster Available.

2024

Edge Detection in Images

École des Ponts

C++ research on perceptual boundary saliency; implemented detection algorithms.

2022

SKILLS

Languages French (native), English (fluent), Spanish (proficient)

Technical Python (PyTorch, Transformers), Docker, Slurm, C++, SQL, R, $\text{\LaTeX}$

ACADEMIC SERVICE

Conference Reviewing Reviewer for CVPR 2026

MISCELLANEOUS

Sports École des Ponts football team; competitive tennis

Interests Passionate about ML research, actively following scientific publications